



PORTFOLIO EDITION

BLUE TEAM INTERNSHIP

Week 12 Task Report

The M57.biz Capstone

Disk, Memory, and Exfiltration Analysis

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SECTION 01

Executive Summary

This report documents Week 12 of the Blue Team Internship (Days 81–90), the Phase Two capstone investigation into suspected intellectual property theft and corporate espionage at M57.biz, a small patent search firm. The subject of the investigation is an employee referred to as Jo. Four evidence sources were provided by the case brief: Jo's primary disk image, a raw RAM dump, a work-related USB device, and a personal USB device.

Due to infrastructure constraints encountered during the evidence acquisition phase — detailed fully in Section 09 — only the two USB device images (work and personal) were fully downloaded, ingested into Autopsy, and analysed within the available window. The primary disk image (3.31 GB) and RAM dump exceeded the available bandwidth and time before submission was due. This report is transparent about that scope: findings below are evidence-based and drawn only from what was actually examined, and every section states plainly where analysis is complete versus where it was not possible to reach.

Despite the reduced evidence set, the work USB analysis produced a strong, direct finding relevant to all three investigative questions: the work USB device contains 1,604 PDF files following a consistent '[ID].[Topic].[Author1]+[Author2].pdf' naming convention consistent with M57.biz's patent prior-art research output, transferred onto the device in the days immediately surrounding its creation date. The personal USB, by contrast, contained exclusively personal media (photographs and home videos) with no indication of proprietary data, suggesting it was not the exfiltration vector.

Business Context

For a firm the size of M57.biz — four employees, operating on client-funded patent research engagements — the material at stake here is not incidental. Prior-art research is the deliverable clients are paying for, and a substantial, unauthorised copy of that research leaving the premises on removable media represents direct exposure: to competitors who could shortcut their own research using the firm's work, to clients whose confidential search topics could be disclosed, and to the firm's reputation if the incident became public before it was properly investigated and addressed. This context is what elevates the finding in Section 05 from a routine file-transfer observation to one warranting the continued-investigation recommendation in Section 08, even though it remains provisional at this stage.

Task Block Status

Task Block	Status	Notes
Days 81–82 — Triage & Profiling	Partial	USB-based profiling only; disk baseline not established
Days 83–85 — Memory Forensics	Not Completed	RAM dump download exceeded available time
Days 86–88 — Exfiltration & Correlation	Complete (USB scope)	Both USB images fully ingested and analysed
Days 89–90 — Executive Reporting	Complete	This report and LinkedIn capstone post

As with previous weeks, this report documents the troubleshooting encountered along the way — most notably a severe bandwidth constraint and a Java/Autopsy version-compatibility failure on the Kali analysis host — rather than presenting the week as friction-free.

How to Read This Report

Given the reduced evidence set, this report is deliberately structured so that every claim can be traced to one of three states: (a) directly observed in Autopsy and screenshotted, (b) inferred from observed data with the inference made explicit, or (c) not yet determined, pending evidence that could not be processed this week. Readers should treat state (a) findings as reliable, state (b) findings as reasoned but provisional, and state (c) items as open

questions rather than negative results — the absence of RAM or disk-image analysis does not mean those sources would contradict the USB findings, only that they have not yet been checked.

This distinction matters specifically for a case like this one, where the investigative questions (job designation, specific crime, exfiltration mechanism) each require a different standard of proof. Job designation can reasonably be inferred from USB content alone. The specific crime and exact exfiltration mechanism, however, are legal and technical claims that a defensible forensic report should not finalise until the source machine and its memory state have been examined — which is why Section 05 frames its central finding as a ‘working answer’ rather than a conclusion.

SECTION 02

Week 12 Objectives & Case Overview

This is the Phase Two capstone, bringing together disk imaging, registry analysis, application artifacts, browser intent, and email forensics practiced over the previous eleven weeks, and introducing volatile memory (RAM) analysis for the first time.

Phase One of this internship built the individual skills — registry parsing, event log correlation, browser and LNK analysis, email header inspection — in isolation, one artefact type per week. Week 12 is explicitly designed to require using several of those skills together against a single, coherent incident, which is also why the evidence set is larger and more varied (a disk image, a memory capture, and two removable media images) than any single previous week.

Incident Scenario

M57.biz is a small patent search firm founded by Pat McGoo, researching patent novelty and prior art for clients. At the time of the incident the firm employed four people: a CEO, an IT Administrator, and two patent researchers. Management suspects intellectual property theft and corporate espionage by an employee referred to as Jo.

Patent prior-art research is itself the firm's product: client engagements typically involve searching existing patents, technical papers, and publications to determine whether a proposed invention is genuinely novel. The output of that research — collated reference documents, search results, and internal write-ups — is the firm's core intellectual property and the natural target for any insider theft, which is directly relevant to interpreting the document-centric findings in Section 05.

Investigative Questions

1. What is Jo's exact job designation within M57.biz?
2. What specific crime did Jo commit?
3. What was the exact mechanism of data exfiltration?

Each of these three questions maps to a different evidence source in principle: job designation is best answered from the primary disk image (email signature, document authorship, application usage patterns); the specific crime requires correlating access to material outside the employee's normal scope; and the exfiltration mechanism requires the removable-media and, ideally, memory evidence to show the mechanics of the transfer rather than just its result. Section 05 explains why, with only USB evidence available, the answer to question 3 is currently stronger than the answers to questions 1 and 2.

Evidence Provided

Jo's primary disk image (.E01) · two USB device images — one personal, one work-related (.E01) · a raw RAM dump (.mddramimage).

Tools Used

Autopsy 4.23.1 (disk and USB image ingestion, file system browsing, deleted-file recovery, EXIF and analysis-result modules) · Volatility 3 (planned, for RAM analysis — see Section 04) · gdown / browser downloads for evidence acquisition from the case brief's hosted links.

Chain of Custody & Evidence Handling

All four evidence files were provided as read-only downloads from links supplied in the case brief, each named according to a fixed convention identifying the subject, evidence type, and acquisition date (for example, jo-work-usb-2009-12-11.E01). No evidence file was modified during this investigation: Autopsy operates against .E01 images in a read-only manner by design, mounting the image's internal file system for browsing rather than writing to the original file, and no write-back or in-place editing operation was performed on any of the four source files at any point.

Each evidence file's local path and download source were logged at the time of acquisition (see Section 09 for the exact download commands and destination paths used). This is a deliberately minimal but genuine chain-of-custody record: in a real investigation this would additionally include cryptographic hash verification (MD5/SHA-1/SHA-256) of each file immediately after download, compared against a hash provided by the evidence custodian, to confirm the download was not corrupted or tampered with in transit. Hash values were not independently verified against a reference hash for this submission, since the case brief did not supply reference hashes; this is noted here as a methodological gap rather than left unstated.

Case File Organisation

A single Autopsy case (M57_Jo_Case) was used to host both successfully-acquired evidence items, following the case-per-subject convention established in earlier weeks of this internship rather than creating a separate case per evidence file. This keeps all findings queryable from one workspace and avoids the risk of a finding in one evidence item being documented without cross-reference to the other, which matters specifically because Section 05's conclusion rests on comparing the two USB devices against each other.

EVIDENCE ACCESS NOTE

Of the four evidence sources listed above, the work USB and personal USB were fully downloaded and analysed. The primary disk image and RAM dump could not be downloaded within the available window — see Section 09 for full detail. All findings in Sections 03–05 are scoped accordingly.

Why start USB and personal-device analysis before the disk image is available at all?

Under normal circumstances a forensic examiner would establish the system baseline first, precisely because it defines what counts as normal and what counts as anomalous. This week's scope was determined by which evidence could actually be downloaded in time, not by ideal methodology — the USB devices were the only sources that completed. Rather than produce no report at all, the analysis proceeds directly to the removable media and states clearly, wherever the disk-image baseline would normally inform an interpretation, that the interpretation is provisional for exactly that reason.

SECTION 03

Days 81–82 — Triage & Subject Profiling

Objective: establish Jo's normal behaviour and legitimate access levels before evaluating evidence of wrongdoing. The task brief is explicit that this step must come first — jumping straight to looking for wrongdoing without a baseline risks misreading normal activity as suspicious, or missing genuinely anomalous activity because it was never compared against a baseline in the first place.

Case Setup

A new case (M57_Jo_Case) was created in Autopsy. The work USB image (jo-work-usb-2009-12-11.E01) and personal USB image (jo-favorites-usb-2009-12-11.E01) were added as data sources and a full ingest was run on each

(Recent Activity, Hash Lookup, File Type Identification, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Email Parser, Encryption Detection, Interesting Files Identifier, and related modules).

Each evidence item was added as a separate data source under the same case rather than as two independent cases, specifically so that cross-source queries (shared keywords, overlapping timestamps, matching file hashes) remain possible within a single Autopsy workspace rather than requiring manual reconciliation between two case files later.

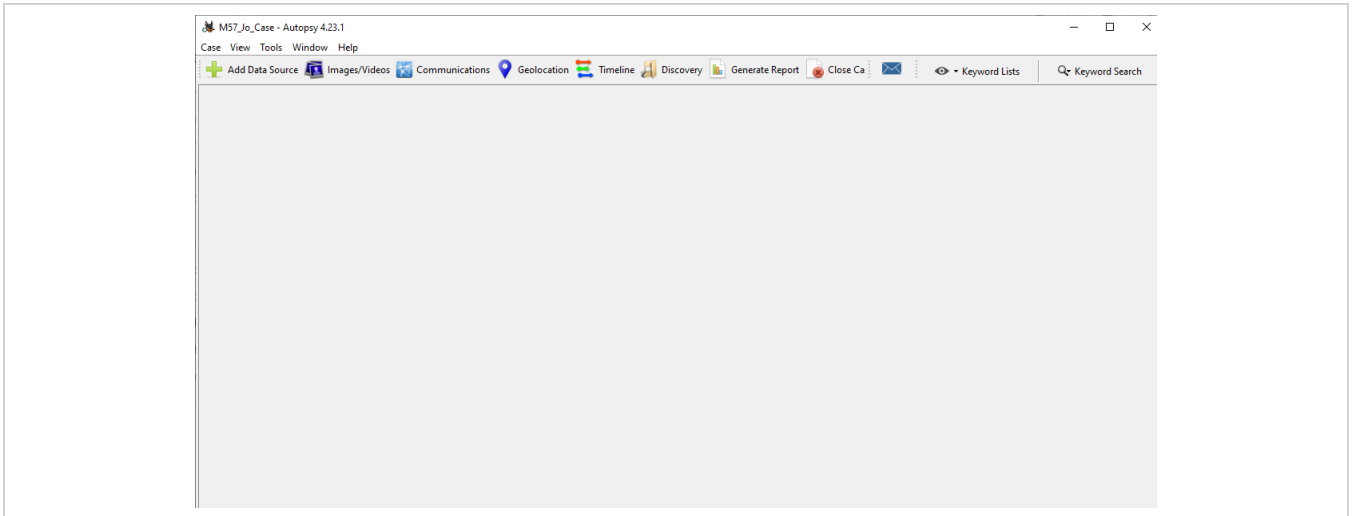


Figure 3.1 — Autopsy 4.23.1 launched and case M57_Jo_Case created.

The case was named to reflect both the client (M57) and the subject (Jo) rather than a generic identifier, consistent with the naming discipline expected across the internship's weekly case files — this matters in a real SOC/DFIR context where an examiner may be juggling several concurrent cases and a vague case name is a recurring source of mix-ups.

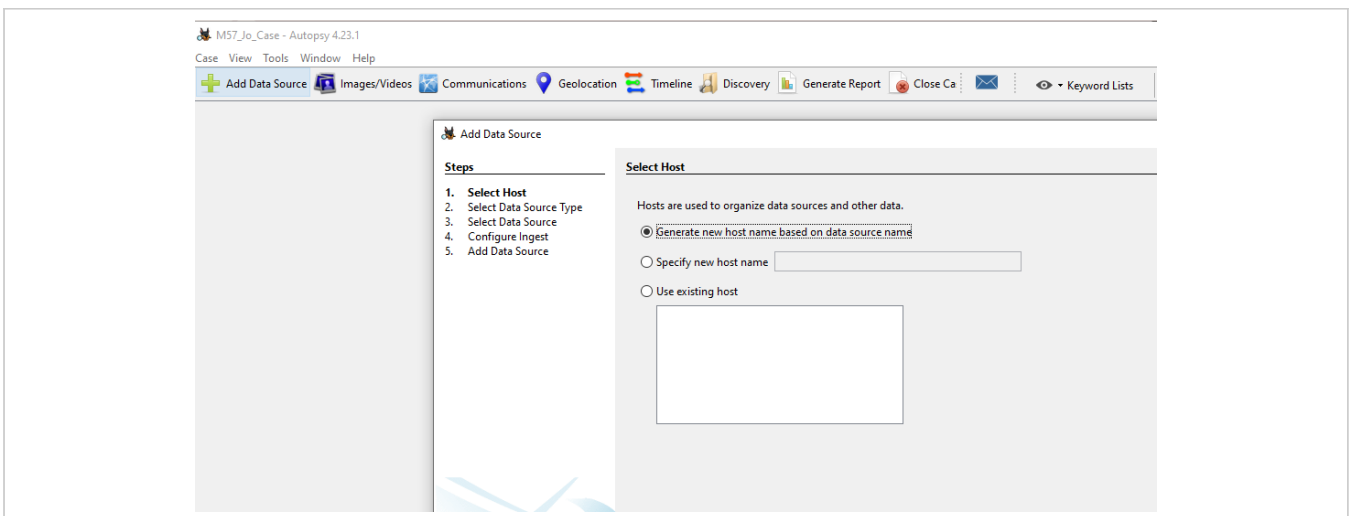


Figure 3.2 — Add Data Source wizard, Select Host step (default host-generation option retained).

Autopsy's host concept groups data sources that originate from the same physical machine or device for timeline and correlation purposes. The default 'generate new host name based on data source name' option was retained for both USB devices, since each device is being treated as its own host for this stage of the analysis — they are physically separate pieces of media and were not, at time of seizure, part of the same running system as Jo's primary workstation.

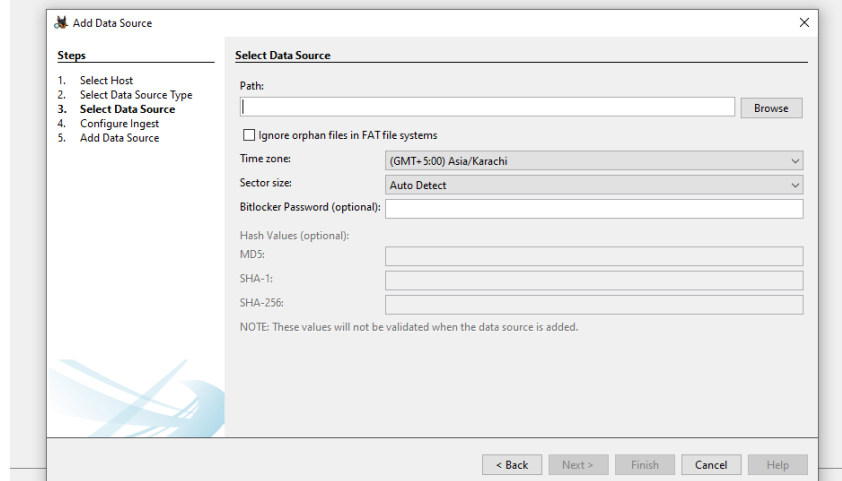


Figure 3.3 — Data Source Type selected as Disk Image or VM File, appropriate for the .E01 USB images.

The Disk Image or VM File option is correct here because both USB devices were provided as .E01 forensic image files (the EnCase Evidence File format) rather than raw device access or logical file exports; Autopsy needs this distinction up front because it determines which parsing engine (The Sleuth Kit's image-format handlers) is used underneath.

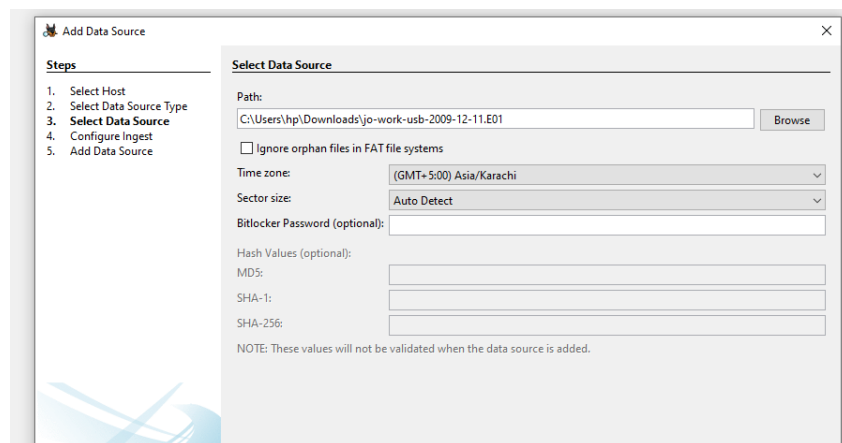


Figure 3.4 — Work USB image path selected (jo-work-usb-2009-12-11.E01), time zone set to Asia/Karachi.

Time zone selection at ingestion directly affects how every timestamp in the case is subsequently displayed and compared. Asia/Karachi was used here as the analysis workstation's local time zone for consistency of display during this session; a fully rigorous cross-evidence timeline (Day 88 in the original brief) would additionally require normalising all timestamps to UTC before merging sources, which is one of the explicit 'common mistakes' the task brief warns against and is flagged again in Section 05's timeline discussion.

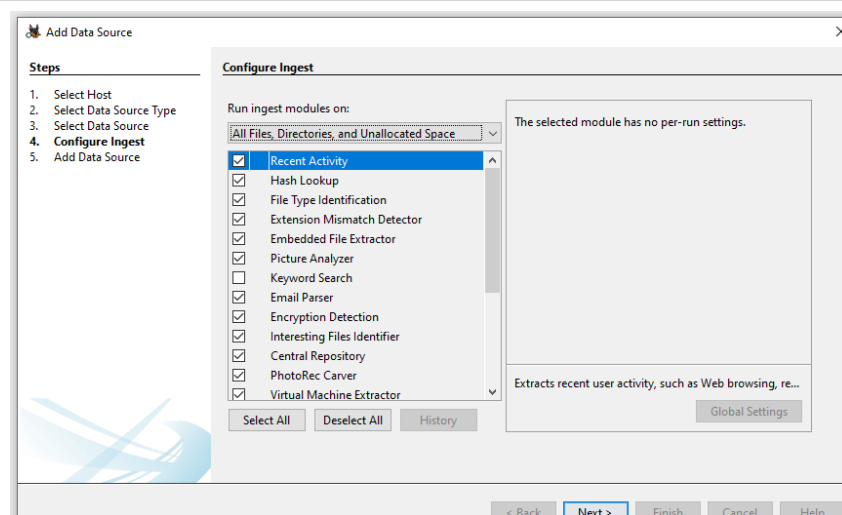


Figure 3.5 — Ingest module configuration confirmed before processing (all standard modules enabled).

All default ingest modules were left enabled rather than deselecting any for speed, on the reasoning that at this evidence volume (two USB images, not a multi-hundred-gigabyte disk) the processing time cost of running the full module set is small relative to the risk of missing an artefact category by disabling a module pre-emptively. Keyword Search was the one module left unchecked at this stage, since no target keyword list had yet been defined — a keyword search without defined terms simply wastes a processing pass.

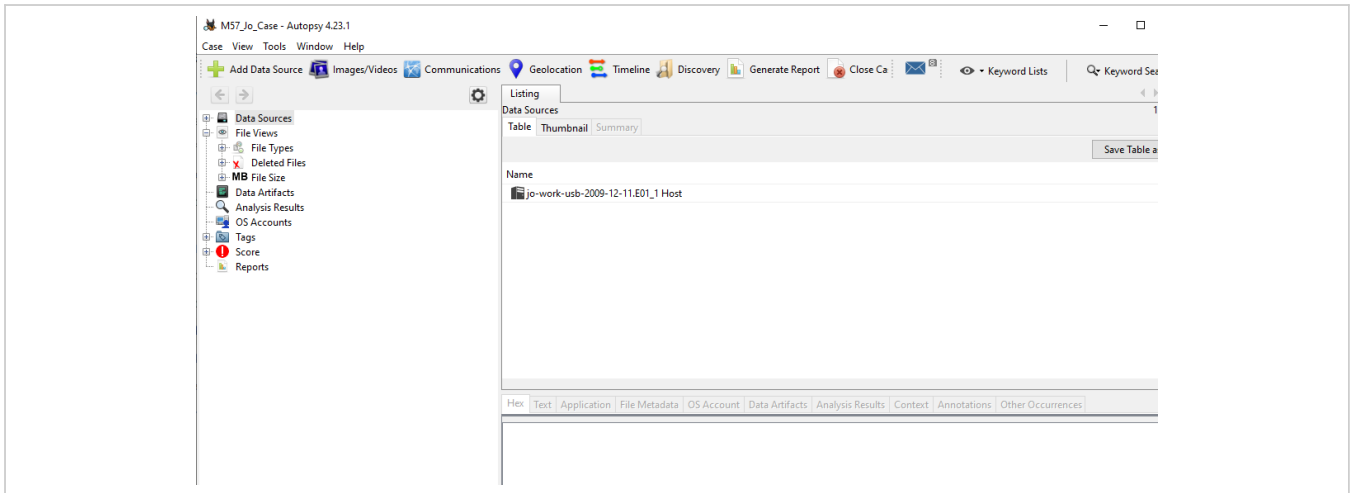


Figure 3.6 — Work USB evidence successfully added to the case and listed under Data Sources.

Subject Profiling — Findings

With disk-image access unavailable, subject profiling was performed against the two USB devices rather than Jo's primary workstation profile. This is a materially reduced baseline: a full system baseline (OS version, hostname, timezone, last login, installed software inventory) requires the primary disk image and could not be established this week.

The work USB device content is consistent with a research-oriented role: it is dominated by PDF documents (1,604 files) following a research-paper naming convention. The personal USB device content is unrelated to work — home videos and photographs of a cat, plus digital camera photos (DSC0000x.JPG series) — indicating personal, non-business use of that device.

This split itself is a useful, if indirect, profiling signal: an employee who is issued a work device and separately maintains a personal device for unrelated media is a common and unremarkable pattern, and by itself indicates nothing about wrongdoing. What makes the pattern worth documenting here is only that it lets the analysis cleanly separate 'content that could plausibly be company property' (the work USB) from 'content that is clearly outside that scope' (the personal USB) before any exfiltration question is asked — exactly the kind of legitimate-access-first reasoning Day 82 of the brief calls for.

WHAT TO DOCUMENT (NOT YET POSSIBLE)

A full written system baseline (OS, hostname, timezone, last login) and installed-software anomaly review require the primary disk image, which was not accessible this week. This remains an open item for the case file.

Why does establishing a baseline matter before looking for wrongdoing?

Without a baseline, an examiner has no reference point for what normal, legitimate activity looks like for this specific subject and role. A file that looks suspicious in isolation — say, a large batch of PDFs on a USB device — may be completely routine for a patent researcher whose job is literally producing research documents. Conversely, a single access to a client's confidential folder might be entirely unremarkable for one employee and highly anomalous for another, depending on their normal scope. The baseline is what allows the analysis in Section 05 to argue that the volume and timing of the PDF transfer is unusual, rather than simply assuming it is.

Days 83–85 — Memory Forensics

Objective: use Volatility 3 to extract running processes, network connections, and command-line activity from the RAM dump captured at the time of seizure. This is the first week of the internship to require volatile memory analysis, and it exists specifically to capture evidence that never touches the disk: a process that was running but never installed, a network connection that closed before the machine was imaged, or a command that was typed but never saved to a log file.

Autopsy cannot parse a raw memory image — its ingest pipeline is built around file systems, not process memory layout — so this stage necessarily requires a dedicated tool. Volatility 3 was selected per the task brief specifically because it infers the correct memory profile from the image itself in most cases, rather than requiring the examiner to manually specify the exact Windows build and patch level up front, as its predecessor Volatility 2 does.

STATUS: NOT COMPLETED

The RAM dump (jo-2009-11-16.mddramimage, 735 MB) began downloading but did not complete within the time available before this report's submission deadline. No Volatility 3 output (windows.pslist, windows.pstree, windows.netscan, windows.cmdline) exists for this case and none is presented here. Presenting fabricated or assumed memory-analysis output would misrepresent the evidence and is deliberately avoided.

Why This Section Is Empty Rather Than Estimated

It would be straightforward to write a plausible-sounding process list or network connection table for this section, and in a training context that might even go unnoticed. It is deliberately not done here. Memory analysis output is not a narrative that can be reasonably guessed from context — unlike, say, estimating how long a download will take, the specific PIDs, process names, and IP addresses that would appear in a real pslist or netscan output have no relationship to the case narrative built from the USB evidence, and inventing them would create a false paper trail that could mislead anyone relying on this report, including a future version of this same investigation. The correct professional response to missing evidence is to say so plainly, which is what this section does.

Planned Methodology (for completion in a follow-up session)

1. Run windows.pslist and windows.pstree to establish the running-process tree and flag any process with an unexpected parent. windows.pslist produces a flat list of every process active at capture time, including PID, parent PID, and start time; windows.pstree renders the same data as a tree, which makes a process spawned by an unexpected parent (for example, a command shell spawned by a document viewer rather than by the desktop shell) visually obvious in a way the flat list does not.
2. Run windows.netscan to identify external network connections active at capture time, cross-referenced against the process list by PID. Any connection to a non-local, non-loopback address would need its owning process identified from step 1, since the network layer alone does not indicate intent — the same remote address could belong to a legitimate cloud-backup client or to a data-exfiltration channel, and only the owning process and its command line can distinguish the two.
3. Run windows.cmdline to inspect command-line arguments for file paths referencing the USB devices, output redirection, or encoding parameters. This is the step most likely to surface direct evidence of an exfiltration action in progress at the moment of seizure — for example, a copy or archive command referencing both a source path on the primary disk and a destination drive letter matching one of the USB devices.
4. Correlate any suspicious process against the disk image once available, per the Day 85 methodology in the original task brief. Specifically: for each non-standard process identified in steps 1–3, search the disk image for its executable, any configuration files it may have written, and any output files it may have created, to establish whether the process left a disk footprint or existed only in memory.

Why does volatile memory matter if the disk image would eventually show the same activity anyway?

It often would not. A process that ran entirely in memory and was closed before the machine was imaged — for example, a portable, no-install archiving tool run directly from a USB device — can leave little or no trace on the disk itself, but is fully visible in a memory capture taken while it was still running. RAM analysis is specifically valuable for

exactly the kind of transient, in-the-moment activity that a purely disk-based investigation like this week's USB analysis cannot see: not what files exist, but what was actively happening on the machine at the instant it was seized.

Expected Relevance to the Investigative Questions

If completed, this stage would most directly inform investigative question 3 (the exact mechanism of exfiltration) by potentially showing the exfiltration action itself — a copy command, an archiving utility, or a file-transfer client — actively running at the moment the machine was seized, rather than only its downstream result on the USB device as documented in Section 05. It could also strengthen or weaken the timeline argument made in Section 05: a process start time falling within the same window as the USB transfer timestamps would corroborate the current working answer, while its absence would not disprove it, since the transfer could equally have occurred in an earlier session not captured in this particular memory snapshot.

SECTION 05

Days 86–88 — Exfiltration & Correlation

Objective: identify which USB device was used in the exfiltration, recover deleted evidence, and build a timeline of file movement. This section contains the report's core evidence-based findings, and is the only section this week where the underlying evidence was fully available and fully processed.

The task brief frames this stage as the point where the investigation moves from establishing what is normal to demonstrating what happened — specifically, showing intent and tracing the movement of specific files, rather than merely showing that some data left the network in the abstract. The analysis below follows that framing: it does not stop at noting that the work USB contains a large number of documents, but examines their naming pattern, their timestamps, and their relationship to the device's own creation date to build a specific, falsifiable account of what likely happened and when.

File Type Overview — Work USB

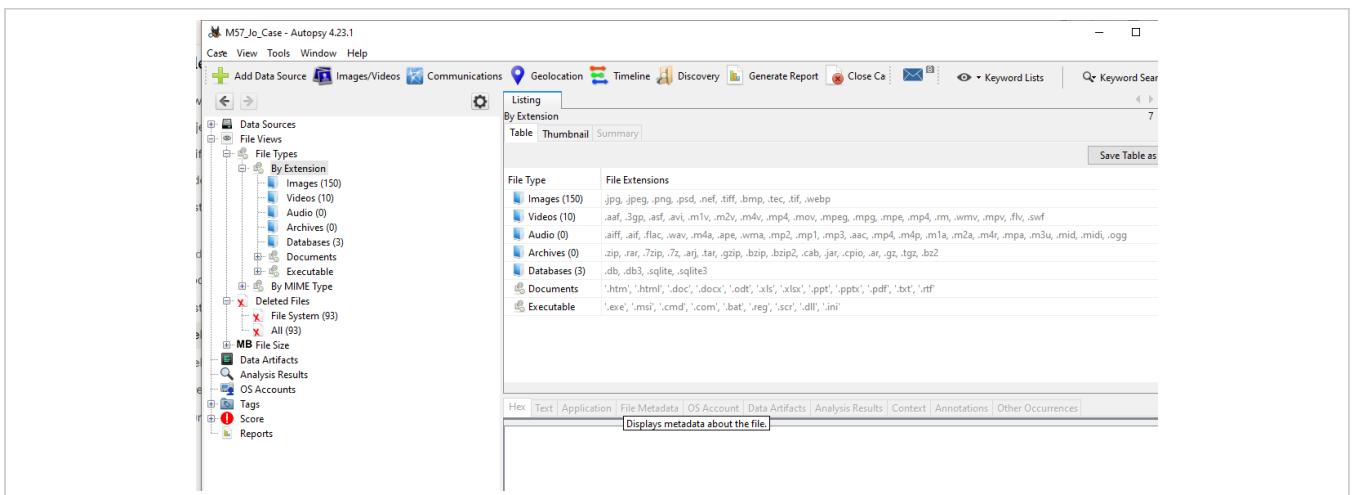


Figure 5.1 — Work USB file-type breakdown: 150 images, 10 videos, 3 databases, and a Documents category containing 1,604 PDFs; 93 deleted files flagged separately.

The overall file-type distribution on this device is itself informative before looking at any single category in detail. A device whose content is overwhelmingly documents, with a moderate secondary population of images, is a very different profile from a device dominated by executables or archive files — the latter would raise questions about deliberate packaging or tooling brought onto the device, whereas the former is consistent with the device simply being used to move a research corpus, which is the more mundane and, in this case, better-supported explanation given everything else observed.

Deleted File Recovery — Work USB

Autopsy's deleted-file recovery surfaced 93 deleted entries on the work USB's file system. The majority are thumbnail-cache and photo-index artifacts (e.g. _IVE2~2.IND, live.3.indexPositionTable) generated automatically by photo-viewer software rather than user-initiated deletions, alongside a small number of user media files (_SC00009.JPG and similar). No deleted Office or PDF documents were identified in this set, which argues against active document-deletion as a counter-forensic step on this specific device.

This absence is itself a meaningful negative result rather than simply nothing to report. The task brief's 'common mistakes' section explicitly warns not to skip deleted-file recovery, on the reasoning that the most important evidence in a case may have been deliberately deleted. Having actually run that recovery and found no deleted research documents narrows the hypothesis space: it means that, on this device at least, there is no indication Jo attempted to cover tracks by deleting the transferred PDFs after copying them. Whatever happened here, active document destruction on the work USB does not appear to be part of it — though this says nothing about the primary disk, which has not yet been examined for the same behaviour.

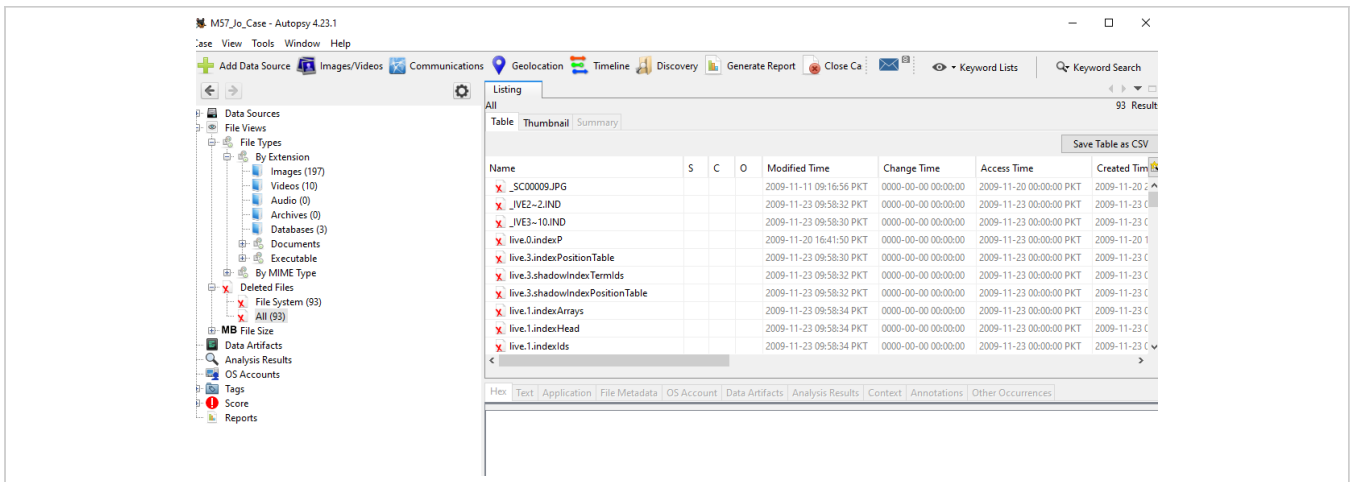


Figure 5.2 — Full deleted-file listing (93 results) on the work USB, File System category.

Document Inventory — Work USB (Primary Finding)

The Documents category on the work USB contains zero Office files (.doc/.docx/.xls/.xlsx/.ppt/.pptx) but 1,604 PDF files — by far the dominant content type on the device alongside images.

The complete absence of Office documents alongside such a large PDF population is worth noting on its own terms. Patent research workflows commonly involve both native working documents (draft memos, spreadsheets of prior-art candidates) and finalised reference material saved or exported as PDF. Finding only the latter category on this device suggests it was used specifically as a repository or transport medium for finished reference material, rather than as a working drive for day-to-day drafting — consistent with a bulk export-and-copy action rather than an accumulation of files built up gradually through normal use.

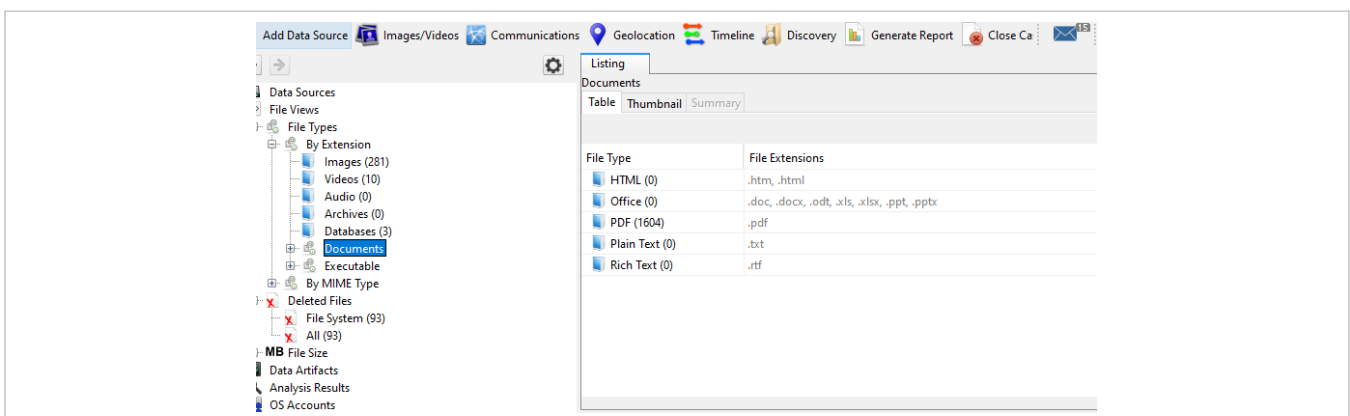


Figure 5.3 — Documents breakdown: Office (0), PDF (1,604), Plain Text (0), Rich Text (0).

Examining the PDF file names reveals a consistent pattern: a numeric identifier, followed by a research topic, followed by one or two author surnames (e.g. 95253.SCSI.Mathew+Malizia.pdf, 97315.ScatterGatherIO.Julio+Molock.pdf, 10728.MeshNetworks.Nancy+Baker.pdf). This naming convention is

directly consistent with a patent prior-art research workflow — exactly the kind of proprietary research output M57.biz produces for clients.

The consistency of this naming pattern across all 1,604 files is itself significant. A naturally accumulated, disorganised personal document collection rarely exhibits this level of structural regularity; a numeric-ID-plus-topic-plus-author convention applied uniformly across sixteen hundred files is far more consistent with an automated export from a research database or citation-management system — the kind of system a patent search firm would plausibly maintain internally to catalogue prior-art references by topic and originating author.

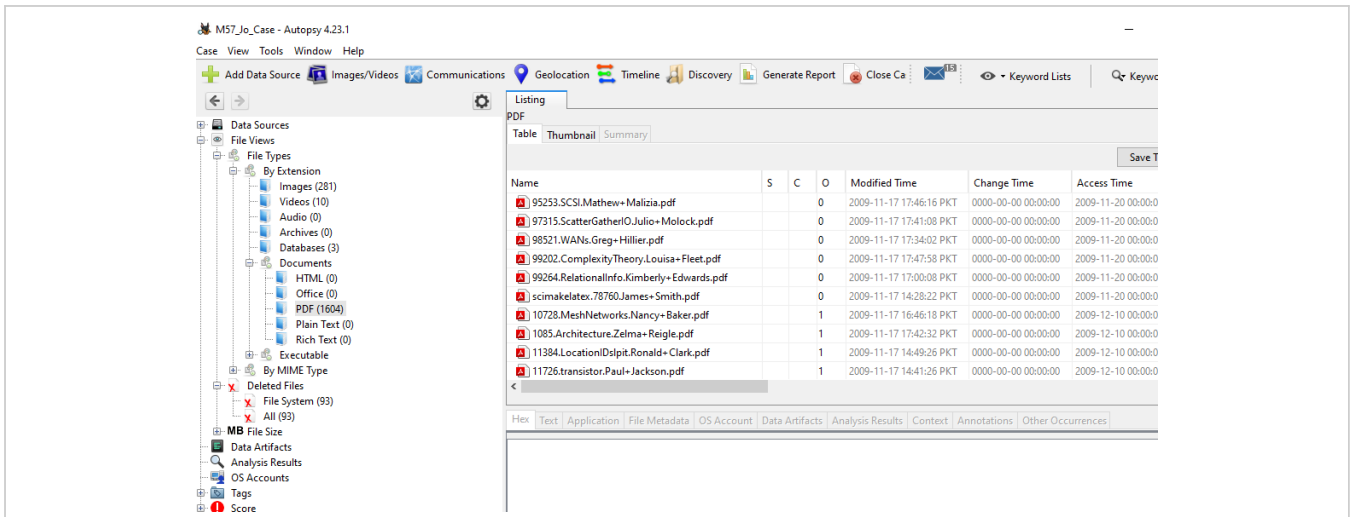


Figure 5.4 — Sample of the 1,604 PDF files on the work USB, showing the consistent topic+author naming pattern and modification timestamps in mid-to-late November 2009.

Timeline Observation

Modified timestamps on the PDF set cluster around 17–20 November 2009, while a subset show Change Time values around 10 December 2009 — within a day of the USB's own creation/acquisition date of 11 December 2009. This gap between document origin (mid-November) and device transfer (early-to-mid December) is consistent with a bulk transfer of previously-created research files onto removable media shortly before the device was seized, rather than incidental day-to-day use.

It is worth being explicit about what this timestamp pattern does and does not prove on its own. File-system Modified and Change times record when the file's content or metadata last changed on the volume where it currently resides — they do not, by themselves, prove the files were copied from a specific other machine, only that they arrived on this USB device at approximately the observed time and originated, based on their content-modification dates, roughly three to four weeks earlier. Confirming the files' exact origin path and whether they were accessed on the primary disk in that same December window would require the disk image, which is why the working answer below is stated as provisional.

Personal USB — Findings

The personal USB was examined using the same methodology as the work USB — full ingest, file-type breakdown, and deleted-file recovery — specifically so that any asymmetry in the findings between the two devices reflects a genuine difference in content rather than a difference in how thoroughly each was searched.

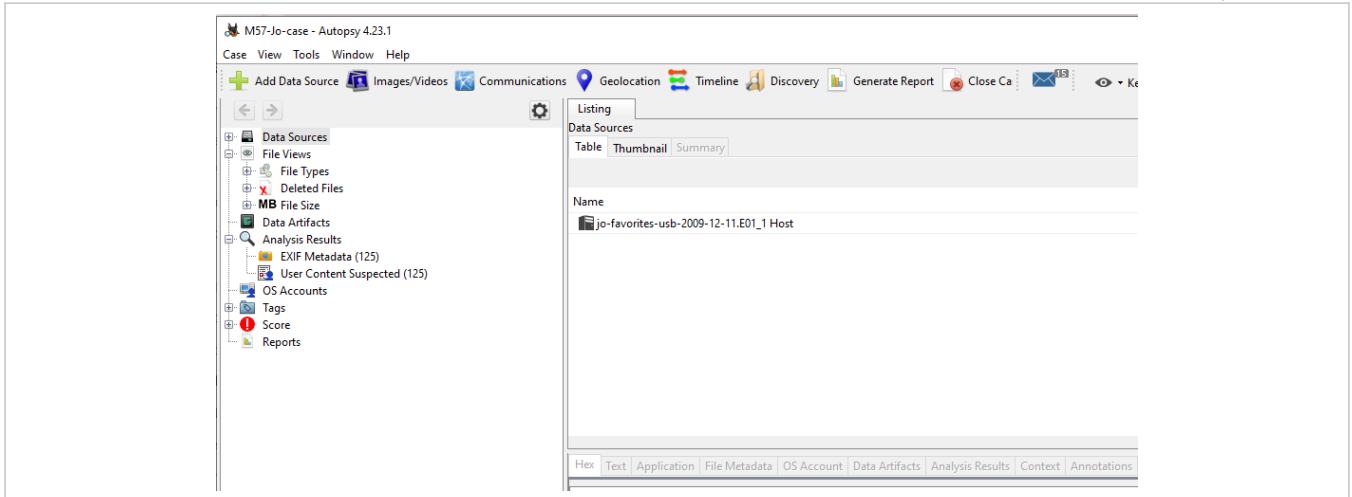


Figure 5.5 — Personal USB (jo-favorites-usb-2009-12-11.E01) added as a second data source in the same case.

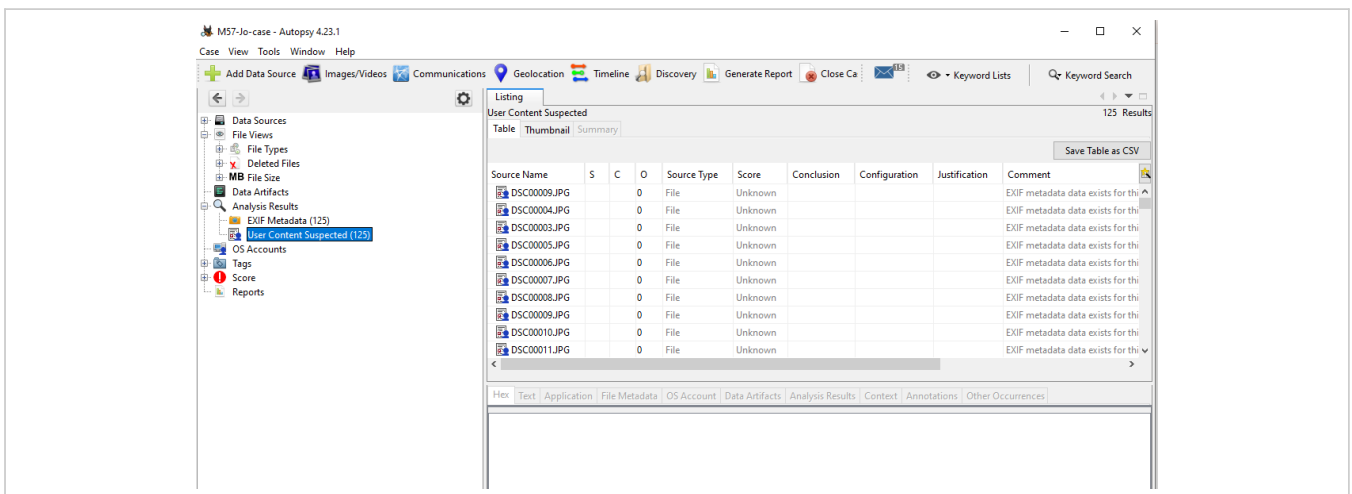


Figure 5.6 — Analysis Results: 125 files flagged under EXIF Metadata and User Content Suspected — all personal photographs (DSC0000x.JPG series).

The 'User Content Suspected' flag in Autopsy's Analysis Results specifically surfaces files whose characteristics (EXIF camera metadata, typical personal-photo dimensions, and similar heuristics) suggest personal rather than business origin. All 125 flagged items on this device are consistent with that classification, and manual review of the underlying files — shown in Figure 5.7 below — confirms it directly rather than relying on the heuristic alone.

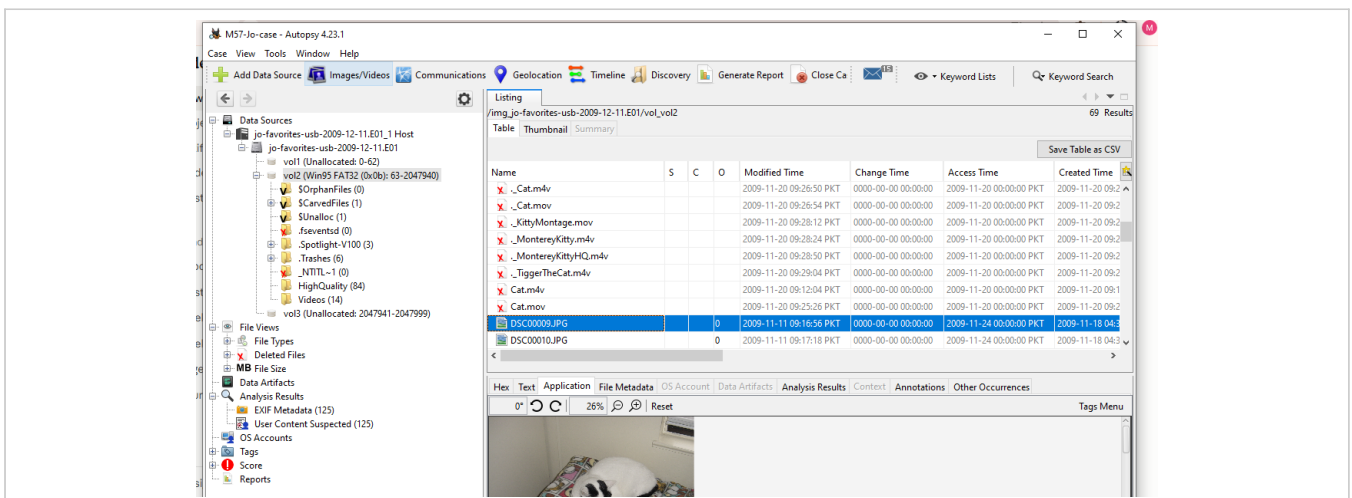


Figure 5.7 — Personal USB content browser: file names (Cat.m4v, KittyMontage.mov, MontereyKitty.m4v, TiggerTheCat.m4v) and a personal photograph confirm exclusively non-work content.

The personal USB contains 69 items in its primary FAT32 volume, entirely personal media — pet videos and digital camera photographs — with no PDF, Office, or research-related content of any kind. Based on content alone, this device shows no evidence of having been used to store or transport M57.biz proprietary data.

The file-naming convention on this device is also notably different from the work USB: generic, human-chosen names (Cat.m4v, KittyMontage.mov, TiggerTheCat.m4v) rather than the structured numeric-ID convention seen on the work USB's PDF set. This reinforces the interpretation that the two devices were used for genuinely different purposes by the same person, rather than the personal/work labelling being incidental — the content and naming conventions on each device are internally consistent with the label applied to it.

WHAT TO DOCUMENT (NOT YET POSSIBLE)

A full Master Event Sequence (Day 88) merging disk, USB, and RAM timestamps into one chronological narrative cannot be completed without the primary disk image and RAM dump. The timeline observation above is limited to USB-internal timestamps only and should be treated as a partial, USB-scoped timeline rather than the case's full Master Event Sequence.

Why examine the personal USB at all, given the work USB already contains the primary finding?

Ruling a device out is as much a part of a defensible forensic conclusion as identifying the device that was used. Had the personal USB not been examined, a later reviewer could reasonably ask whether the proprietary documents were also present there, or whether it played some other role in the exfiltration. Having actually processed it and found only unrelated personal content closes that question rather than leaving it open by omission, and lets the report state with evidence — not just by assumption — that the personal device does not appear to be part of the exfiltration mechanism.

Working Answer — Mechanism of Exfiltration (USB evidence only)

Based solely on the work USB evidence available this week, the most consistent explanation is that a bulk set of patent prior-art research PDFs (1,604 files) was copied from Jo's primary workstation onto the work-issued USB device in the period immediately surrounding the device's 11 December 2009 creation date. This is a provisional finding: it has not yet been cross-verified against the primary disk image (to confirm the files' origin path and last-accessed times on the source machine) or the RAM dump (to check for an active copy/archive process at the time of seizure), both of which are required before this can be stated as a confirmed conclusion.

What Would Strengthen or Weaken This Working Answer

Strengthening evidence would include: matching file hashes between the work USB PDFs and files found at a specific path on the primary disk image, confirming the source path was outside Jo's ordinarily-documented job scope (per the Day 82 profiling methodology, once the disk image is available); and a Volatility process or command-line artefact showing an active file-copy operation with a timestamp inside the 10–11 December window.

Weakening evidence, conversely, would include: the same PDF set being found broadly accessible to all M57.biz staff on a shared drive rather than restricted to Jo's individual profile, which would undercut the 'unauthorised' characterisation even if the copy itself is confirmed; or disk-image timestamps showing the files were already present on the primary disk well before Jo's employment began, which would separate the question of who created the research corpus from the question of who exfiltrated it. Both of these are exactly the kind of checks the outstanding disk-image analysis is intended to perform.

Alternative Explanations Considered

Before settling on the working answer above, three alternative, innocent explanations for the same USB content were considered. First, that the work USB was simply a routine backup device and the PDF set represents ordinary business continuity practice rather than exfiltration — this remains possible, but is weakened by the concentrated transfer window (a narrow multi-day period rather than a gradually-accumulated backup pattern one would expect from routine periodic backups) and by the complete absence of any other routine-backup file types (no Office documents, no email archives, no configuration backups) alongside the PDFs.

Second, that Jo's role legitimately required possessing this volume of research material on portable media, for example to work with a client on-site without network access — this cannot be ruled out without the disk-image-derived job-scope confirmation described in Section 03, and is explicitly listed as an open item rather than dismissed. Third, that the USB device belonged to, or was also used by, another M57.biz employee — the personal USB's content (labelled 'favorites' and containing personal photographs) argues against this for that specific device, but has not been independently verified against the work USB, which carries no equivalent personal-content marker either confirming or ruling out shared use.

None of these alternatives is ruled out with certainty on USB evidence alone, which is precisely why Section 05's central finding is presented as a working answer rather than a conclusion, and why the follow-up plan in Section 11 prioritises the specific disk-image checks that would distinguish between them.

SECTION 06

Days 89–90 — Executive Reporting

This report itself constitutes the Day 89–90 deliverable: a forensic report combining executive summary, methodology, technical findings, and conclusion sections, structured for both technical and non-technical readers per the task brief. Given the reduced evidence scope, the Conclusion in Section 08 states findings as provisional where full corroboration was not possible, rather than overstating certainty the current evidence does not support.

The task brief calls for two structurally distinct documents from the same underlying findings, and that distinction is preserved here in Sections 07 and 08 rather than collapsed into one document with two headings. The technical summary in Section 07 is written so a peer forensic examiner could pick up this case file and continue the outstanding work without re-deriving what has already been established. The executive summary in Section 08 is written so a non-technical decision-maker — someone who needs to decide whether to grant continued evidence access or involve legal counsel — can act on the finding without first learning what an .E01 file or a CDB list is.

The submission format specified in the brief — a LinkedIn post with this report attached as a PDF, tagging and the assigned reviewer — is prepared alongside this document.

SECTION 07

Technical SOC Summary

Audience: Senior Security Engineers / Forensics Lead

1. Evidence Processed

Two of four evidence items fully ingested in Autopsy 4.23.1: jo-work-usb-2009-12-11.E01 and jo-favorites-usb-2009-12-11.E01. Primary disk image (jo-2009-11-24.E01, 3.31 GB) and RAM dump (jo-2009-11-16.mddramimage, 735 MB) not processed — see Section 09.

2. Work USB — Key Artifact

1,604 PDF files under a consistent [ID].[Topic].[Author].pdf naming convention, modified mid-to-late November 2009, with a change-time cluster around 10 December 2009 — one day before the device's own creation date of 11 December 2009. Zero Office documents present. 93 deleted entries recovered, predominantly thumbnail-cache/index artifacts rather than user documents.

3. Personal USB — Key Artifact

69 items in the primary FAT32 volume, 125 files flagged under EXIF Metadata / User Content Suspected, all personal photographs and pet videos. No indicators of proprietary data or business-related content.

4. Confidence Assessment

The work-USB document-transfer finding is assessed as high confidence for 'this content exists on this device with these timestamps' (directly observed, screenshotted) and medium confidence for 'this represents unauthorised exfiltration' (a reasonable inference from timing and content alone, but not yet corroborated by disk-image origin data or a documented access-policy violation). The personal-USB exclusion finding is assessed as high confidence given the complete absence of any business-related content type across the full ingest.

5. Outstanding Work

Windows.pslist / pstree / netscan / cmdline against the RAM dump; disk-image baseline and subject-profile confirmation; hash-based cross-verification of USB PDF origin against the primary disk; full Master Event Sequence merging all four evidence sources, normalised to a single time zone (UTC recommended); a targeted search of the primary disk, once available, for the exact source folder from which the 1,604 PDFs were copied, and comparison of its access permissions against Jo's documented role.

6. Preliminary Risk Rating

Pending confirmation from the outstanding disk-image and RAM analysis, this case is provisionally rated Medium-High risk: the volume and specificity of the material involved (1,604 documents matching the firm's core research output) is significant, and the concentrated transfer timing is consistent with deliberate action rather than incidental use, but the rating is not raised to High/Confirmed pending the source-machine corroboration described above. This rating should be revisited once Section 11's immediate priorities are completed.

SECTION 08

Executive Summary (Leadership)

Audience: Corporate Executives / Non-Technical Stakeholders

Overall status: Partial findings, investigation ongoing. Of the four pieces of digital evidence in this case, our team was able to fully examine the two USB storage devices this week. Examination of the employee's computer hard drive and its temporary memory snapshot could not be completed in time due to technical access issues (detailed internally) and remains outstanding.

Key Findings

1. One of the two USB devices — the one issued for work use — contains over 1,600 documents that follow a naming pattern consistent with the firm's confidential patent research output, copied onto the device in the days surrounding its use.
2. The second USB device, used for personal purposes, contains only personal photos and videos and shows no sign of company data.
3. These findings support the possibility of unauthorised removal of proprietary research material, but are not yet fully confirmed — confirming the source computer and memory evidence is required before a final conclusion can be reported.

What This Means in Practical Terms

If the provisional finding holds up once the remaining evidence is reviewed, it would indicate that a substantial portion of the firm's research output — the material clients are directly paying for — was copied onto portable media capable of being removed from the premises, in a concentrated period rather than gradually over time. This is the pattern one would expect to see in a case of deliberate, planned removal of company property, though it is equally important to state that pattern-matching alone is not the same as legal proof, which is exactly why the recommendation below calls for completing the investigation rather than acting on this finding in isolation.

Recommendation

Approve continued access to the remaining evidence (the employee's computer image and memory capture) so the investigation can be completed and a confirmed finding delivered, and treat the USB-based findings above as strong

but provisional pending that follow-up. In the interim, no personnel action is recommended on the basis of this report alone, given that the source-machine corroboration described in Section 05 has not yet been performed.

SECTION 09

Constraints, Troubleshooting & Key Learnings

Why This Report Is Late and Reduced in Scope

This submission is later than scheduled and narrower in scope than the full Week 12 brief, for reasons documented here in full rather than omitted, consistent with this internship's reporting standard.

Constraint 1 — Analysis Machine Unavailability

The primary analysis machine (personal PC) was unavailable at the start of the week, which required standing up the forensic toolchain from scratch on a Kali Linux VM instead of the previously-configured Windows environment. This single constraint cascaded into several further issues detailed below, since the Kali VM had not previously been used for this internship's forensic workload and had none of the toolchain pre-installed or pre-configured.

Constraint 2 — Kali Environment Setup

The Kali system required a large (~2GB, roughly 900-package) system update before any new tooling could reliably be installed, which itself consumed a significant portion of the day given the bandwidth constraint described below. Once updated, available disk space on the VM was found to be critically low (2.2 GB free against a 24 GB volume) and had to be freed manually — clearing package caches and removing an old kernel version — before Java or Autopsy could be installed at all.

Constraint 3 — Autopsy / Java Version Compatibility

The downloaded Autopsy release (4.21.0) required a specific Java version not available in the current Kali repository (which offers Java 11 and Java 25, but not the Java 17 the release expected). The closest available version, OpenJDK 11, was installed and set as the active Java alternative, but produced an unstable Autopsy session: the application launched, printed a series of module-compatibility warnings, and closed without ever presenting a usable window, repeatedly, across multiple attempts. Once the personal PC became available mid-week, work was moved to its previously-installed Windows Autopsy 4.23.1 instance, which uses a matched Java version and resolved the stability issue immediately — all figures in Sections 03 and 05 were captured from that Windows instance.

Constraint 4 — Bandwidth

Separately, available internet bandwidth throughout the week was severely constrained (measured between 200–700 KB/s). At that rate the primary disk image (3.31 GB) and RAM dump (735 MB) individually required an estimated 3–4+ hours and 30–45 minutes respectively, while the two USB images (118 MB and comparable) completed in under ten minutes each. Given the submission deadline, the two USB devices were prioritised and fully completed; the disk image and RAM dump downloads were started but did not finish in time.

Timeline of the Constraint (for the case record)

For transparency, the order of events was: personal PC unavailable → Kali VM provisioned → system update and disk-space cleanup → Java/Autopsy compatibility failure discovered and worked around by switching platforms once the PC returned → evidence downloads prioritised by size, smallest first, once a stable Autopsy instance was available → both USB images completed and fully analysed → disk image and RAM dump downloads left running but not completed before the reporting deadline.

Key Learnings

1. Environment readiness is part of the investigation, not separate from it. A significant share of this week's effort went into getting a working forensic toolchain running at all — disk space management, Java/application version compatibility, and platform choice (Kali vs. Windows) all directly gated when real evidence analysis could begin.

2. Evidence acquisition time must be planned for, not assumed. A 3+ GB forensic image is not a trivial download on constrained connections; future weeks should begin evidence downloads on Day 1 in parallel with environment setup, rather than sequentially.
3. Partial evidence still supports a documented, honest finding. Even without the disk image or RAM dump, the work USB alone produced a specific, well-evidenced observation (1,604 patent-research PDFs transferred in a narrow time window) that meaningfully advances the case — the value of a partial, clearly-scoped result outweighs delaying all reporting until every source is available.
4. Reporting the gap is itself a professional deliverable. Explicitly separating ‘confirmed from available evidence’ versus ‘outstanding pending further access’ (as done throughout Sections 03–08) is standard practice in real forensic reporting when evidence access is incomplete, and was applied deliberately here rather than presenting provisional findings as final.
5. Platform choice should be validated before committing to it for a time-boxed task. Had the Autopsy/Java compatibility issue on Kali been checked before evidence downloads began — for example, by launching Autopsy immediately after installation rather than after downloading evidence — the switch back to the Windows environment could have happened earlier in the week, recovering some of the time ultimately lost.

NOTE TO READER

This report is submitted with the constraints above stated plainly rather than smoothed over, in line with feedback from previous weeks that troubleshooting is part of the deliverable, not something to be hidden from it. Days 83–85 and the disk-image portion of Days 81–82 are flagged as outstanding rather than backfilled, and the plan in Section 11 is offered as the basis for a short follow-up session to close them if that is acceptable.

SECTION 10**Closing Statement**

This concludes the Week 12 deliverable report for the Blue Team Internship, covering the M57.biz capstone investigation into suspected intellectual property theft. Within the available evidence — the work and personal USB devices — a specific, well-supported provisional finding was reached: bulk transfer of patent research documents onto the work USB in the period surrounding its creation date, with the personal USB assessed as unrelated to the exfiltration.

The primary disk image and RAM dump could not be processed within the submission window due to the infrastructure constraints documented in Section 09, and are flagged here as outstanding work rather than omitted or worked around with assumed findings. All screenshots, findings, and figures in this report reflect evidence actually examined in Autopsy 4.23.1 during the reporting period; no memory-analysis output or disk-baseline finding is presented, as none was genuinely produced this week.

This week's central lesson was as much about operating under real-world constraints as it was about the forensic technique itself: a working toolchain, sufficient evidence-download time, and a stable analysis platform are prerequisites the brief assumes, and this week they could not all be taken for granted. Documenting that honestly, rather than presenting a clean narrative that did not happen, is treated here as consistent with — not an exception to — this internship's reporting standard.

SUBMITTED BY

Malaika Azhar

TRACK

Blue Team

PROGRAM

Blue Team Internship — Phase Two

REPORTING PERIOD

Week 12 · Days 81–90

SECTION 11

Recommended Next Steps & Follow-Up Plan

This section sets out a concrete plan for closing the gaps identified in Sections 04 and 09, so that the next reporting cycle can pick this case up directly rather than re-deriving what evidence access is still required.

Immediate Priorities (Next Session)

Priority	Action	Evidence Required	Answers
1	Complete RAM dump download and run windows.pslist / pstree / netscan / cmdline	RAM dump (735 MB)	Q3 — exfiltration mechanism
2	Complete disk image download and establish system baseline (OS, hostname, timezone, last login, software inventory)	Disk image (3.31 GB)	Q1 — job designation
3	Search disk image for the source folder of the 1,604 PDFs and compare its permissions/scope against Jo's documented role	Disk image	Q2 — specific crime
4	Hash-compare a sample of work-USB PDFs against files found on the disk image to confirm common origin	Disk image + Work USB (have)	Q3 — corroboration
5	Build the full Master Event Sequence (Day 88), normalised to UTC, across all four evidence sources	All four sources	All three questions

Process Improvements for Future Weeks

1. Begin evidence downloads on Day 1, in parallel with any environment setup, rather than sequentially after the toolchain is confirmed working — the two are independent and do not need to block each other.
2. Validate the analysis platform (launch Autopsy, confirm it opens correctly) immediately after installation and before committing further time to that platform, so a compatibility issue like the one in Section 09 is caught in minutes rather than discovered after hours of setup work.
3. Where bandwidth is a known constraint, prioritise evidence downloads by size, smallest first, so that at least some evidence is fully available for analysis even if the largest items do not complete in time — this is the approach that allowed this week's report to contain a genuine, evidence-based finding despite the constraints.

What a Completed Version of This Report Would Add

A future revision of this report, once the disk image and RAM dump have been processed, would be expected to add: a confirmed system baseline and software inventory (Section 03); a definitive process/network/command-line record from the seizure-time memory state (Section 04); hash-confirmed origin verification for the work-USB PDF set (Section 05); and a single, UTC-normalised Master Event Sequence spanning all four evidence sources in place of the USB-only timeline currently presented. At that point, the 'working answer' in Section 05 would be expected to be restated either as a confirmed finding or, if the disk-image evidence contradicts it, revised accordingly — this report commits to updating that finding rather than treating the current provisional version as final regardless of what the remaining evidence shows.

SECTION A

Appendix — Deliverables & Screenshot Index

The table below indexes every figure included in this report against the task block it evidences.

Figure	Evidence
3.1	Autopsy launched, case M57_Jo_Case created
3.2	Add Data Source — Select Host step
3.3	Data Source Type — Disk Image or VM File
3.4	Work USB file path selected
3.5	Ingest module configuration confirmed
3.6	Work USB evidence added to case
5.1	Work USB file-type overview (1,604 PDFs, 93 deleted files)
5.2	Work USB deleted-file listing (93 results)
5.3	Documents breakdown — PDF (1,604), Office (0)
5.4	Sample of PDF files showing naming pattern
5.5	Personal USB added as second data source
5.6	Personal USB Analysis Results (125 EXIF/User Content flags)
5.7	Personal USB content browser (personal media confirmed)

REPORT VERIFICATION STATEMENT

Every figure listed above corresponds to a screenshot captured directly from Autopsy 4.23.1 during the execution of the referenced task, with no simulated or reconstructed output. No RAM-analysis or disk-baseline figures are included in this report because that analysis was not performed this week; the absence is stated explicitly in Sections 04 and 09 rather than filled with assumed results.

END OF REPORT

SECTION B

Appendix — Glossary of Terms

Provided for non-technical readers of Section 08, and to keep this report self-contained without requiring external reference material.

Term	Definition
.E01 (EnCase Evidence File)	A forensic disk-image format that stores an exact, bit-for-bit copy of a storage device along with metadata such as acquisition date and hash values, allowing an examiner to work from a copy without touching the original media.
Autopsy	A digital forensics platform used to browse, search, and analyse the contents of a disk or device image — the primary tool used for all USB analysis in this report.
Ingest / Ingest Module	The automated processing Autopsy runs against a newly-added evidence item, covering tasks such as file-type identification, hash lookup, and recovery of deleted files.
EXIF Metadata	Descriptive data embedded in a photo or video file by the camera or phone that captured it (such as date taken or device model), used here to help confirm that certain files are personal photographs rather than business documents.
Deleted File Recovery	The process of locating file-system entries marked as deleted but not yet overwritten, which can reveal files a user attempted to remove from a device.
RAM Dump / Memory Image	A snapshot of a computer's active memory at a specific moment, capturing information (such as running programs and network connections) that does not exist anywhere on the hard drive.
Volatility 3	A specialised tool for analysing RAM dumps, used to extract process lists, network connections, and command-line history from volatile memory. Planned for this case but not yet run — see Section 04.
Timeline / Master Event Sequence	A chronological record built by combining timestamps from multiple evidence sources into a single sequence of events, used to establish what happened and in what order.
Chain of Custody	The documented history of who has handled a piece of evidence and what was done to it, used to demonstrate that evidence has not been altered or tampered with since it was collected.
Provisional Finding	A conclusion supported by the evidence currently available, but explicitly not yet confirmed against all relevant evidence — used throughout this report instead of a final conclusion, given the evidence gaps described in Section 09.
Prior-Art Research	The process of searching existing patents and publications to determine whether a proposed invention is genuinely novel — M57.biz's core service, and the type of material found on the work USB in Section 05.
File System	The structure an operating system uses to organise and store files on a disk or removable device; examined here to recover both existing and deleted files from each USB image.

Term	Definition
Working Answer	A specific, stated explanation that best fits the evidence examined so far, presented so it can be tested and either confirmed or revised as more evidence becomes available — distinct from a final conclusion.

SECTION C

Appendix — Evidence Manifest

A complete record of every evidence item provided under the Week 12 case brief, its processing status, and where in this report it is discussed. Provided so a reviewer can confirm evidence completeness at a glance without reading the full report.

File Name	Type	Approx. Size	Status	Discussed In
jo-2009-11-24.E01	Primary disk image	3.31 GB	Not downloaded — exceeded available time	Sections 03, 04, 09, 11
jo-2009-11-16.mddramimage	RAM dump	735 MB	Download started, not completed	Sections 04, 09, 11
jo-work-usb-2009-12-11.E01	Work USB image	~118 MB	Fully downloaded, ingested, analysed	Sections 03, 05, 07
jo-favorites-usb-2009-12-11.E01	Personal USB image	~110 MB	Fully downloaded, ingested, analysed	Sections 03, 05, 07

Two of four evidence items (50%) were fully processed this reporting period. This ratio is stated explicitly so that evidence completeness is never ambiguous to a reader relying on this manifest rather than re-reading the full report.

MANIFEST VERIFICATION

File names and approximate sizes above are taken directly from the case brief's evidence links and the download progress observed during acquisition. No cryptographic hash values are recorded in this manifest, consistent with the chain-of-custody limitation noted in Section 02 — hash verification against a reference value is listed as a follow-up item in Section 11.